

Unit 3, Lesson 2: Quotient Laws of Exponents



Benchmark

MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.



Learning Target

- ☐ I can explain and use the quotient of powers, power of a quotient, and zero exponent laws.



3.2.1 Warm-Up: Use a Pattern

Look at the pattern shown in the figures below. Work with your partner to answer the questions.



Figure 1

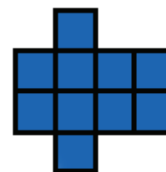


Figure 2

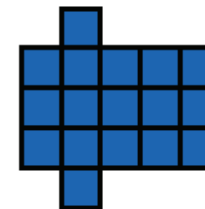


Figure 3

- Describe the pattern in words.
- If the pattern continued, how many squares would be in Figure 10?



3.2.2 Activity: Station 1: Quotient of Powers Exponent Law

Work with your partner or a small group to complete three stations involving the laws of exponents.

Task 1:

- Review the steps Rosa, Victor, and Emma each used to generate an equivalent expression. Then, answer the questions that follow.

	Rosa	Victor	Emma
Given	$\frac{5^6}{5^3}$	$\frac{10^4}{10^2}$	$\frac{7^4}{7^1}$
Student Work	$\frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5}$ $5 \cdot 5 \cdot 5$	$\frac{10 \cdot 10 \cdot 10 \cdot 10}{10 \cdot 10}$ $\frac{\boxed{10}}{\boxed{10}} \cdot \frac{\boxed{10}}{\boxed{10}} \cdot \frac{10}{1} \cdot \frac{10}{1}$ $1 \cdot 1 \cdot 10 \cdot 10$ $10 \cdot 10$	$\frac{7 \times 7 \times 7 \times 7}{7}$ $7 \times 7 \times \frac{\boxed{7}}{\boxed{7}} \times 7$ $7 \times 7 \times 7$
Equivalent Expression	5^3	10^2	7^3

- What do you notice about the bases of the numerator and denominator of each given exponential expression?

- Circle the exponents in the given exponential expression and circle the exponent in the equivalent expression each student created. How are they related?

Task 2:

- Fill in the blank to complete the rule for dividing exponential expressions with the same bases.



When dividing two powers that have the same nonzero base, keep the base and _____ the exponents. This is called the **quotient of powers exponent law**.

- When a is a nonzero real number, and m and n are integers, the exponential expression $\frac{a^m}{a^n}$ is equal to

- ☐ a^{mn} .
- ☐ a^{m-n} .
- ☐ a^{m+n} .

Task 3: Show your steps of applying the quotient of powers exponent law to evaluate.

- $\frac{(-2)^6}{(-2)^3}$
- $\frac{(3)^6}{(3)^2}$
- $\frac{(-7)^4}{(-7)^1}$





3.2.3 Activity: Station 2: Power of a Quotient Exponent Law

Task 1: Luis and Maya have shown their steps for using the power of a quotient exponent law to create equivalent expressions. Amir is confused. Help Amir complete his task by showing similar steps to Luis and Maya for his expression $\left(\frac{7}{3}\right)^4$.

Luis	Maya	Amir
$\left(\frac{8}{10}\right)^3$	$\left(\frac{3}{11}\right)^5$	$\left(\frac{7}{3}\right)^4$
$\frac{8}{10} \cdot \frac{8}{10} \cdot \frac{8}{10}$	$\frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11}$	$\frac{7}{3} \cdot$
$\frac{8 \cdot 8 \cdot 8}{10 \cdot 10 \cdot 10}$	$\frac{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3}{11 \cdot 11 \cdot 11 \cdot 11 \cdot 11}$	
$\frac{8^3}{10^3}$	$\frac{3^5}{11^5}$	

Task 2:

- Fill in the blank to complete the rule for raising a fraction to a power.



When raising a fraction to a power, the result is equal to the quotient of the numerator and denominator raised separately to the _____ power. This is called the **power of a quotient exponent law**.

- When a and b are nonzero real numbers, and m is an integer, the exponential expression $\left(\frac{a}{b}\right)^m$ is equal to

- ☐ a^{b-m} .
- ☐ $\frac{a^m}{b^m}$.
- ☐ $a^m - b^m$.

Task 3: Apply the power of quotient exponent law. Do not simplify the fractions.

- $\left(\frac{-4}{8}\right)^2$

- $\left(\frac{-4}{-2}\right)^6$

- $\left(\frac{2}{-8}\right)^5$



3.2.4 Activity: Station 3: Zero Exponent Law

Task 1: Two students used their understanding of exponents to write an equivalent expression for the exponential expression $\frac{2^3}{2^3}$.

1. Discuss each step that the students completed.

Hannah	Sebastian
$\frac{2^3}{2^3}$	$\frac{2^3}{2^3}$
$\frac{2 \times 2 \times 2}{2 \times 2 \times 2}$	2^{3-3}
$\frac{2}{2} \times \frac{2}{2} \times \frac{2}{2}$	2^0
$1 \times 1 \times 1$	
1	

2. Using Hannah's and Sebastian's work, make a prediction about the value of any number raised to the power of 0.

Task 2: Use the tables to answer the questions.

Exponent Form	2^4	2^3	2^2	2^1	2^0
Value	16	8	4	2	

Exponent Form	3^4	3^3	3^2	3^1	3^0
Value	81	27	9	3	

1. What pattern do you see in the top row of each table?
2. What pattern do you see in the bottom row of each table?
3. How is the pattern in the bottom row of each table related to the value of the base in the top row of the table?
4. Use the patterns in the tables to find the values of 2^0 and 3^0 . Fill in the empty cell in each table.

$$2^0 = \underline{\quad}$$

$$3^0 = \underline{\quad}$$

Task 3: Fill in the blank to complete the rule for a base with a zero exponent.



When a nonzero real number is raised to the zero power, a^0 , its value equals _____. This is called the **zero exponent law**.



3.2.5 Bring It Together: Exponential Expressions with Rational Bases

1. Match each set of expressions with the Exponent Law that it demonstrates. Use arrows to indicate matches.

$\left(\frac{0.5}{1.9}\right)^3$ $\left(\frac{0.5}{1.9}\right) \cdot \left(\frac{0.5}{1.9}\right) \cdot \left(\frac{0.5}{1.9}\right)$ $\frac{0.5 \cdot 0.5 \cdot 0.5}{1.9 \cdot 1.9 \cdot 1.9}$ $\frac{0.5^3}{1.9^3}$	
$\left(\frac{2}{3}\right)^0$ 1	
$\frac{10.1^6}{10.1^4}$ $\frac{10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1}{10.1 \cdot 10.1 \cdot 10.1 \cdot 10.1}$ $\frac{10.1}{10.1} \cdot \frac{10.1}{10.1} \cdot \frac{10.1}{10.1} \cdot \frac{10.1}{10.1} \cdot \frac{10.1}{1} \cdot \frac{10.1}{1}$ $1 \cdot 1 \cdot 1 \cdot 1 \cdot 10.1 \cdot 10.1$ 10.1^2	

Zero Exponent Law

Quotient of Powers Law

Power of a Quotient Law

2. Complete the following steps for each expression.

- Rewrite each expression shown by using the laws of exponents.
- Check off the exponent law used from the list.
- Evaluate the exponential expression.

$\frac{0.5^8}{0.5^5}$	☐ quotient of powers law ☐ power of quotient law ☐ zero exponent law
$\frac{0.5^8}{0.5^5} = \underline{\hspace{2cm}}$	

$\left(\frac{0.2}{10}\right)^4$	☐ quotient of powers law ☐ power of quotient law ☐ zero exponent law
$\left(\frac{0.2}{10}\right)^4 = \underline{\hspace{2cm}}$	

$\left(\frac{3}{4}\right)^0$	☐ quotient of powers law ☐ power of quotient law ☐ zero exponent law
$\left(\frac{3}{4}\right)^0 = \underline{\hspace{2cm}}$	



3.2.6 Your Turn!

1. Rewrite the expressions using laws of exponents. State the exponent law used.

Expression	Equivalent	Exponent Law
$\frac{8.6^6}{0.2^6}$		
$\frac{15.7}{15.7}$		
$\frac{6.75^7}{6.75^3}$		
$\left(\frac{5}{20}\right)^{10}$		
$\left(\frac{0.85}{2.64}\right)^7$		
8.45^0		



3.2.7 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about using the quotient law of exponents.